

Machine Learning Engineering Bootcamp

Capstone Project: Pick Your Deployment Method and Engineering Plan

Step 9

Learning Objective

- Understand that deploying a ML model is not the same as training one, and that a lot of other considerations need to be accounted for
- Learn the various popular options available to a ML Engineer through all the standard open source technologies as well as top cloud providers (AWS, Google, Azure, IBM)
- Learn the various engineering and MLOps skills necessary to deploy and run a ML Model in a production setting
- Learn early that “deployment” is just the beginning of the life cycle for your model, and how important it is to integrate in your plan what happens next, and what you will do as a MLE to care and monitor your model once it is making live predictions.

Criteria	Meets Expectations	Weightage
Time Estimate	2 - 4 Hours	
Completion	❑ A clear and concise deployment plan (1 point) ❑ Plan goes beyond deployment and includes caring for the model once deployed, how to monitor it, and redeploy it after retraining (1 point)	2 points
Process and understanding	❑ Demonstrating understanding of the various options available for the problem at hand (2 points)	6 points

	❑ Weighing various factors such as costs, speed of deployment, performance of the model once deployed, options to monitor the model once deployed, logging, etc.. (2 points) ❑ Demonstrating how it fits with the rest of the ML pipeline (for data processing, model training, tuning, etc.. (2 points)	
Presentation	❑ Google Doc or Slides (1 point) ❑ Pseudo code / diagrams to support engineering and deployment plan (1 point)	2 points

Excellence:

1. *Student is not using any of the premade deployment options on any of the cloud providers, but instead writes from scratch his own deployment, packaging and infrastructure (e.g. Kubernetes) for their model. Alternatively, students demonstrate a high level of engineering throughout their whole plan, showing a production-level mentality.*